

EE 508: Fundamentals of Electric Drives

Course Code	EE 508
Course Name	Fundamentals of Electric Drives
L-T-P-C	3-0-0-3
Prerequisites	EE 309 (Power Electronics), EE 201 (Electromechanics)
Intended for	UG/PG
Distribution	Elective for EE
One-time approval	recommended by the 9th BoA (Oct 2015)

Course Contents

- **Introduction to drives:** [2 Lectures]
 - Importance of drives, Factors governing the choice of drives, Basics of drive dynamics, Types of load, Selection of motor power rating, Applications
- **DC motor drives:** [17 Lectures]
 - Types of dc motors, starting and braking, transient analysis of separately excited motor with armature and field control [4 Lectures]
 - Controlled rectifier fed drives, multi-quadrant operation of separately excited dc motor fed from fully-controlled converter [4 Lectures]
 - Control of electric drives, closed loop torque, speed and position control, current and speed sensing [5 Lectures]
 - Chopper controlled dc drives [2 Lectures]
 - Supply harmonics, power factor and current ripple [2 Lectures]
- **Induction motor drives:** [18 Lectures]
 - Induction motor operation with non-sinusoidal voltage supply, starting and braking of induction machines [4 Lectures]
 - Methods of speed control stator voltage control, variable frequency control, field weakening [4 Lectures]
 - Voltage Source Inverter (VSI) control of induction motors [3 Lectures]
 - Current Source Inverter (CSI) control of induction motors [3 Lectures]
 - Current regulated VSI control [2 Lectures]
 - Introduction to Vector Control [2 Lectures]
- **Synchronous motor drives:** [5 Lectures]
 - Types of synchronous motors Cylindrical-rotor and Salient-pole motors, Operation from fixed frequency supply starting, braking, load disturbance transients
 - Variable frequency control true-synchronous mode, Self-controlled mode, Self-controlled
 - Synchronous motor drive using Load Commutated Thyristor Inverter
 - Introduction to special machines Permanent Magnet synchronous motor, Brushless dc motor etc.

Text Books

1. G. K. Dubey, **Fundamentals of Electric Drives**, Alpha Science International Ltd., 2001.
2. Dubey G. K., **Power Semiconductor Controlled Drives**, Prentice Hall International Edition. 1989.

References

1. Mohan N., Undeland T.M. and Robbins W.P., **Power Electronics Converters, Applications and Design**, 3rd Edition, Wiley India. 2008
2. Bose B. K., **Power Electronics and Variable Frequency Drives Technology and Applications**, IEEE Press, Standard Publisher Distributors, 2001
3. Rashid M., **Power Electronics - Circuits, Devices and Applications**, 3rd Edition, Pearson Education, 2003.
4. Krause, P. C., Wasynczuk, O., Sudhoff, S. D., **Analysis of Electric Machinery and Drive Systems**, Wiley-Interscience, 2002.
5. S. K. Pillai, **A First Course on Electrical Drives**, New Age International Pvt. Ltd., 2004
6. R. Krishnan, **Electric Motor Drives: Modeling, Analysis, and Control**, Prentice Hall, 2001.
7. N. K. De and P. K. Sen, **Electric Drives**, Prentice-Hall of India Pvt. Ltd., 1999